

Ismael Hernandez

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SUMMARY

Robotics Engineering student with hands-on experience in software development, backend/frontend applications, and system integration. Strong foundation in object-oriented programming, algorithms, and problem-solving

HARD SKILLS

Git

Github

Python

JavaScript

SQL

APIs

Web development

SOFT SKILLS

Problem-solving, Proactivity, Responsibility, Curiosity

RELEVANT PROJECTS

Robotic Control and Simulation Platform

- Developed a robotics control platform integrating React, Python, ESP32 hardware, and real-time WebSocket communication.
- Implemented generalized forward and inverse kinematics algorithms for configurable serial manipulators.
- Created interactive 3D visualization tools using Three.js and React Three Fiber to monitor robot state and coordinate systems.
- Designed a modular robot configuration framework allowing robot structure, kinematics, and UI controls to be generated from JSON definitions.

Boat Rental Management System

- Developed a web application to manage boat rental operations, fleet information, and payment records.
- Designed and implemented CRUD workflows for maintaining operational and business data through an intuitive user interface.
- Delivered an MVP adopted in a real business environment and refined features based on operational feedback.

EDUCATION

B.Sc. Robotics Engineering

Universidad de Guadalajara · En curso